Contact

www.linkedin.com/in/sahar-malakshah-a6778029 (LinkedIn)

Top Skills

R (Programming Language)
Analytical Skills
Research Computing

Certifications

Advanced Learning Algorithms
Machine Learning Specialization

Publications

Joint variant and de novo mutation identification on pedigrees from high-throughput sequencing data

Joint variant and de novo mutation identification on pedigrees from high-throughput sequencing data

Sahar Malakshah

Bioinformatics Scientist with ML/AI experience in Genomics & Proteomics
Union City, California, United States

Summary

Accomplished Bioinformatics Scientist with over a decade of industry experience in leading-edge genomics, bioinformatics, and machine learning applications. Proven ability to lead cross-functional projects, particularly in cancer genomics and diagnosis, human genomics, rare diseases, and epigenetics.

Expertise spans NGS and Omics data analysis, bioinformatics pipeline development, skilled in developing and utilizing machine learning models and AI-driven tools for predictive analytics, FDA regulatory communications for product approval, and deep understanding of NGS assay and wet lab process.

Demonstrates strong programming and data analysis skills as well as molecular biology and omics lab experience, provides unique perspective in working in Genomics field to drive advancements in healthcare.

Experience

ATUM
Senior Bioinformatics Scientist
January 2026 - Present (4 months)
Newark, CA

Independent Research Project
Genomics Research Project — ML-Guided Epigenomics Target Reprogramming in Cancer (GitHub)
June 2025 - October 2025 (5 months)
San Francisco Bay Area

Built a multi-modal ML pipeline integrating DNA methylation and RNA-seq data to identify possible cancer therapeutic targets. Used PCA dimensionality reduction and DMR reversal for interpretability and target prioritization.

GRAIL

4 years 11 months

Staff Bioinformatics Scientist

August 2022 - October 2024 (2 years 3 months)

Menlo Park, California, United States

Senior Bioinformatics Scientist

December 2019 - August 2022 (2 years 9 months)

Working closely with assay team to manage the data produced in the lab and AI team to make an understanding of the lab data.

Fabric Genomics

Senior Bioinformatics Scientist

September 2016 - November 2019 (3 years 3 months)

Mostly involved in designing, developing, bench-marking of companies new products in clinical diagnosis by analysis of large and complex genomics data sets including cancer data; Leveraging machine learning and statistical methods to generate predictions and biological insights

Affymetrix

Bioinformatics Scientist II

March 2014 - August 2016 (2 years 6 months)

Santa Clara

Real Time Genomics

Bioinformatics Scientist

February 2013 - January 2014 (1 year)

BINA Technologies

Bioinformatics Engineer

October 2012 - December 2012 (3 months)

Redwood city, CA

I just finished my internship at Bina Technologies (a fast-growing startup company in Bay Area) in which scalable software and hardware systems and high performance computing platforms for analysis of high-throughput human genome data is developping.

Here is the focus of my research there:

*Integrated analysis of human Next Generation Sequencing data with scripting/pipeline designs (using Perl & BioPerl modules, R & Bioconductor packages and also shell script for analysis & graphics) and relevant open

source tools (e.g. BWA, Seqalto, Soap, Novoalign, Samtools, Picard, GATK, IGV, etc.)

Education

University of California, Riverside

Ph.D, Evolutionary Genomics, Computational Biology

University of Tehran

M.S, Biotechnology

University of Tehran

B.S, Biotechnology